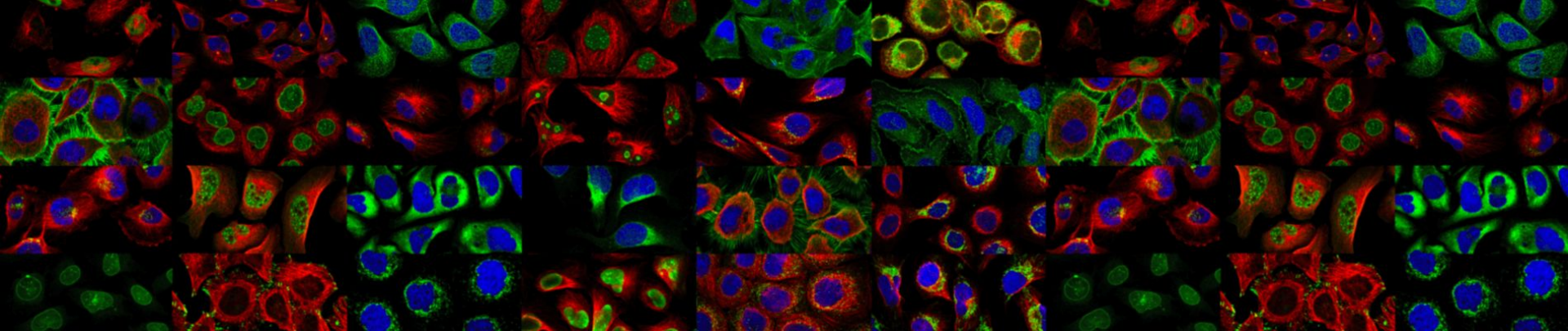


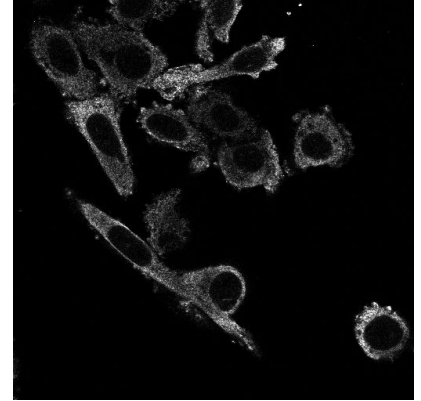
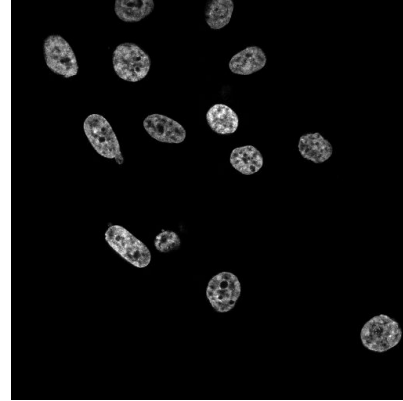
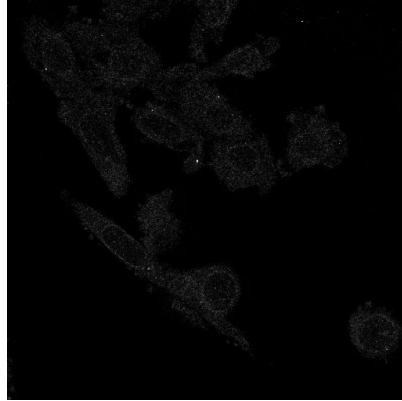
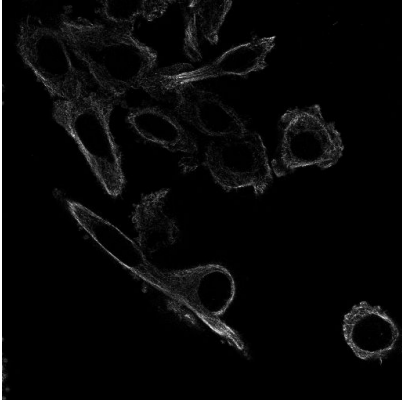


Unveiling Cellular Secrets: The Human Protein Atlas Image Classification Challenge

Dataset: Human Protein Atlas Image Classification
Niklas Hohn, Christopher Arleth

The Challenge

Classify the locations of proteins
within a cell from microscope images



The Labels

28 labels

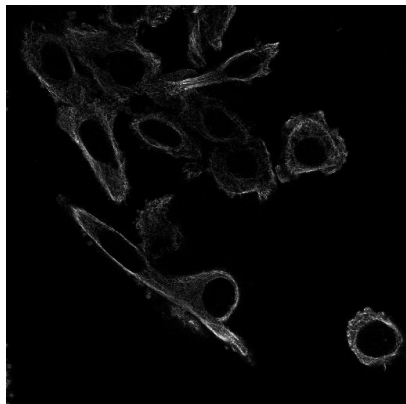
- | | |
|------------------------------|-----------------------------------|
| 0. Nucleoplasm | 14. Microtubules |
| 1. Nuclear membrane | 15. Microtubule ends |
| 2. Nucleoli | 16. Cytokinetic bridge |
| 3. Nucleoli fibrillar center | 17. Mitotic spindle |
| 4. Nuclear speckles | 18. Microtubule organizing center |
| 5. Nuclear bodies | 19. Centrosome |
| 6. Endoplasmic reticulum | 20. Lipid droplets |
| 7. Golgi apparatus | 21. Plasma membrane |
| 8. Peroxisomes | 22. Cell junctions |
| 9. Endosomes | 23. Mitochondria |
| 10. Lysosomes | 24. Aggresome |
| 11. Intermediate filaments | 25. Cytosol |
| 12. Actin filaments | 26. Cytoplasmic bodies |
| 13. Focal adhesion sites | 27. Rods & rings |

The Images

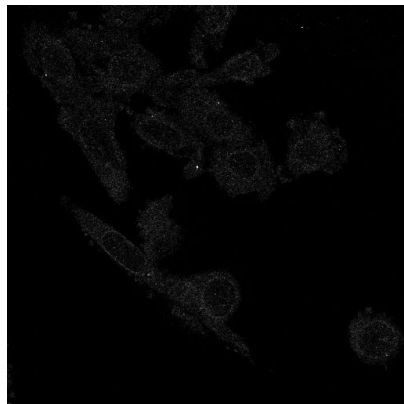
31072 Samples

124288 Images

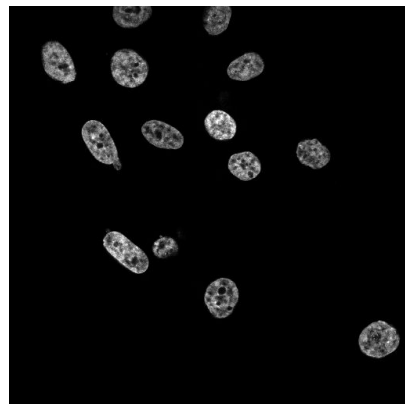
512x512 Size



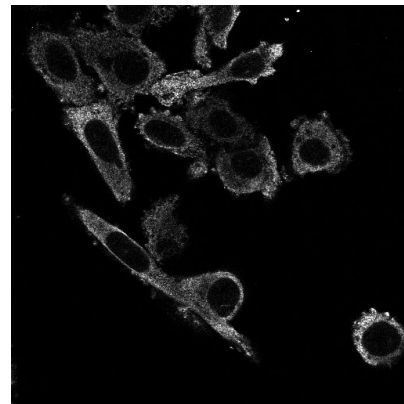
Microtubules



Target Protein

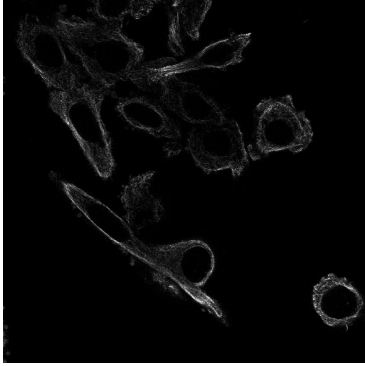


Nucleus

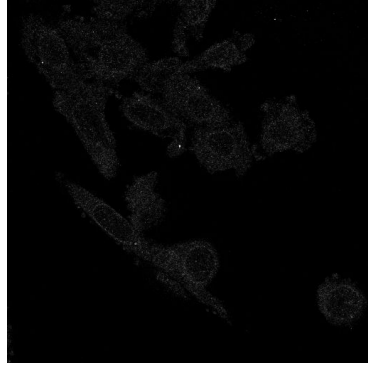


Endoplasmic
reticulum

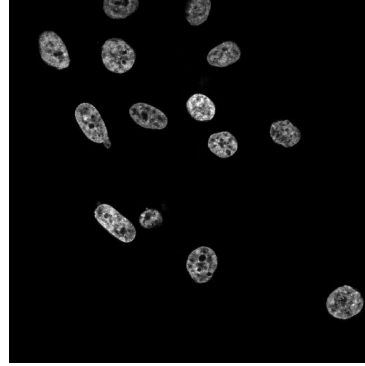
The Images



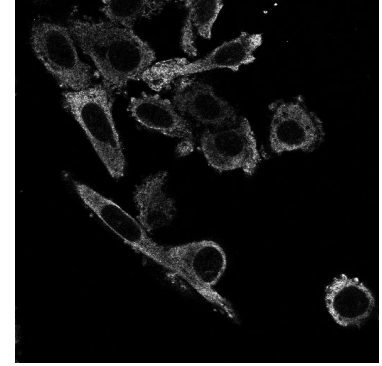
Microtubules



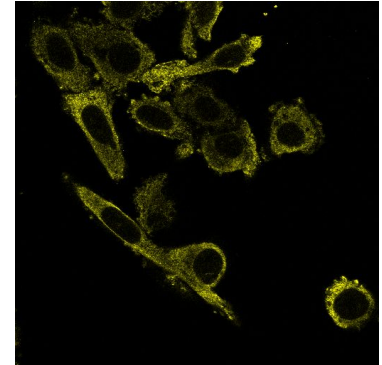
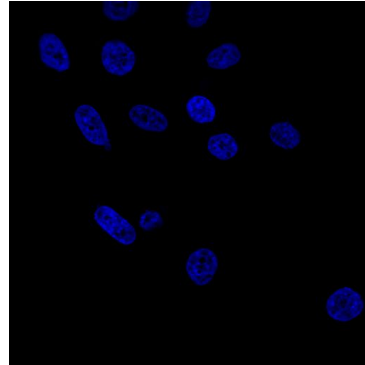
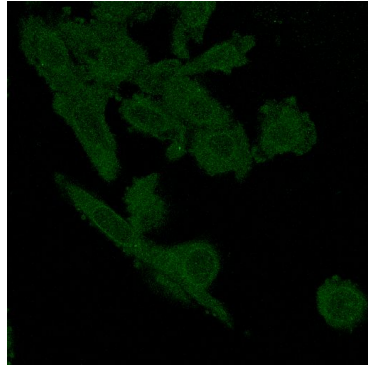
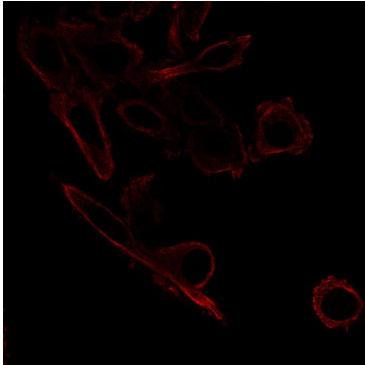
Target Protein



Nucleus

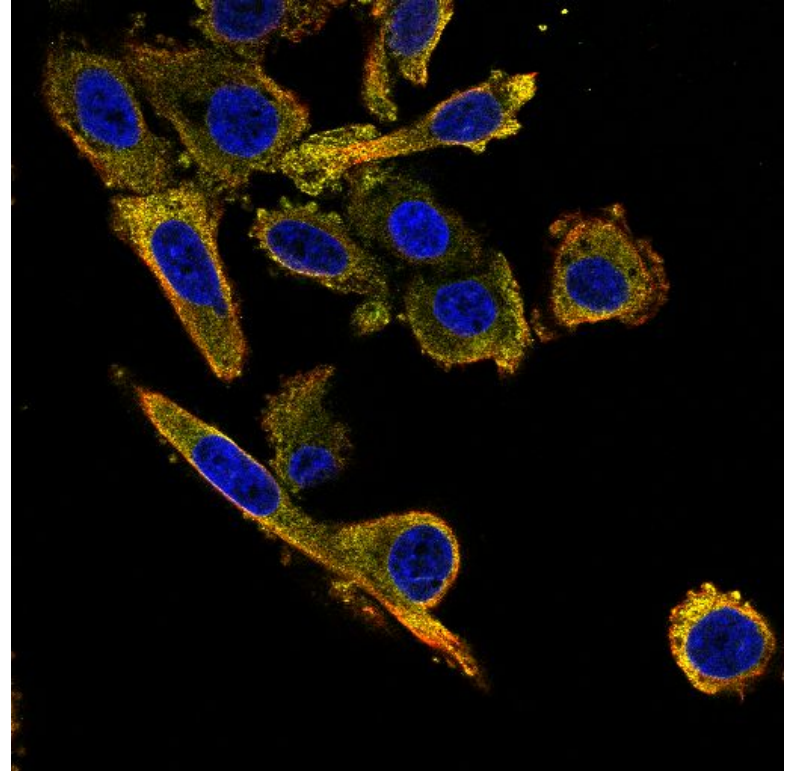
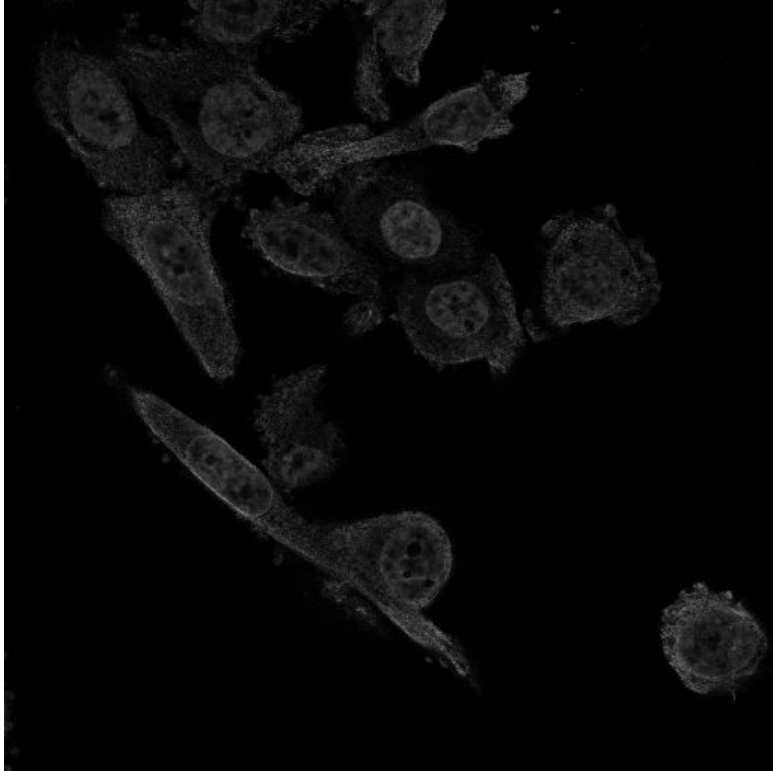


Endoplasmic
reticulum



The Images

Combined

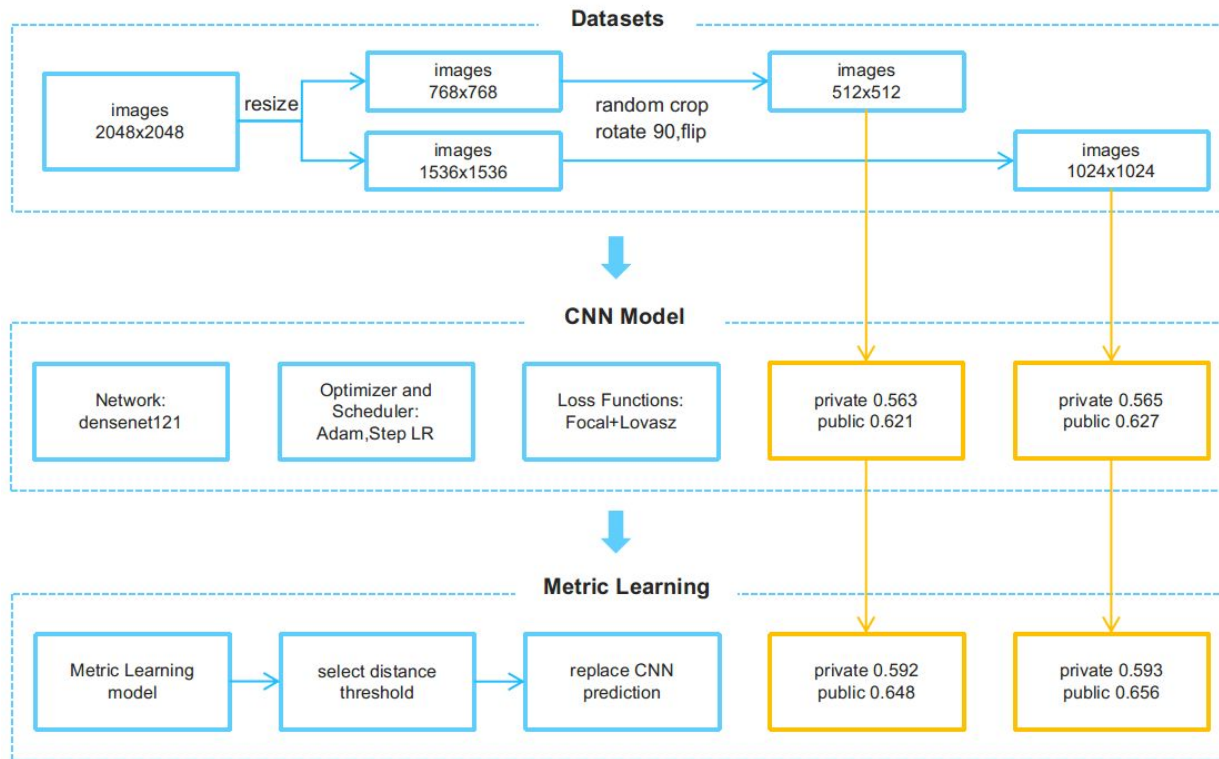


Winner of 2019 Challenge

Team: bestfitting

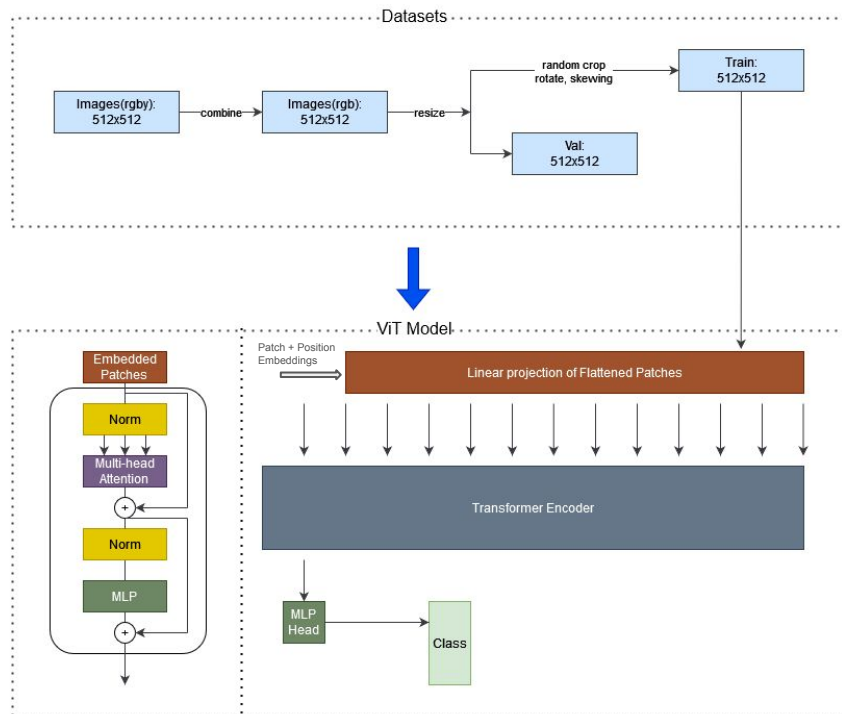
Model: CNN

F1: 0.6956



Our approach

Model: ViT F1: 0.744



Basemodel: vit_base_patch16_384

12 Transformer blocks

21000 Classes

trained on 14 mio images

Hyperparameters:

drop_rate: 0.1

drop_path_rate: 0.1

learning_rate: 0.0005 - 0.00001

batch_size: 128

gradient_clipping_norm: 0.4

What is a ViT?

A powerful deep learning model for computer vision that adapts the architecture originally designed for analyzing text.

Core Concept:

- Images as a Sequence
- Self-Attention Mechanism
- Flexibility
- High Performance

How it Works

Grid of fixed-size patches. Patches are treated like words in a sentence.

Weights the importance of every patch relative to all other patches.

Often requires very large datasets, but can learn complex patterns

Performs exceptional when trained on sufficient data

Simplified Architecture: Input Image → Split into Patches → Transformer Encoder → Classification

ViT vs. CNN

A comparison of two powerful deep learning architectures for computer vision.

Vision Transformer (ViT)

- **Remarkable Performance:** outperforming contemporary CNNs.
- **High Efficiency:** requires fewer computational resources for pre-training compared to CNNs.
- **Architecture:** Based on the transformer model, treats an image as a sequence of patches,
- **Weaker Inductive Bias:** fewer built-in assumptions about the data. highly flexible but depends on large datasets or significant data augmentation to perform well

Convolutional Neural Network (CNN)

- **Strong Inductive Bias:** built with inherent biases (like translation equivariance and locality) that are well-suited for image tasks.
- **Effective with smaller datasets**

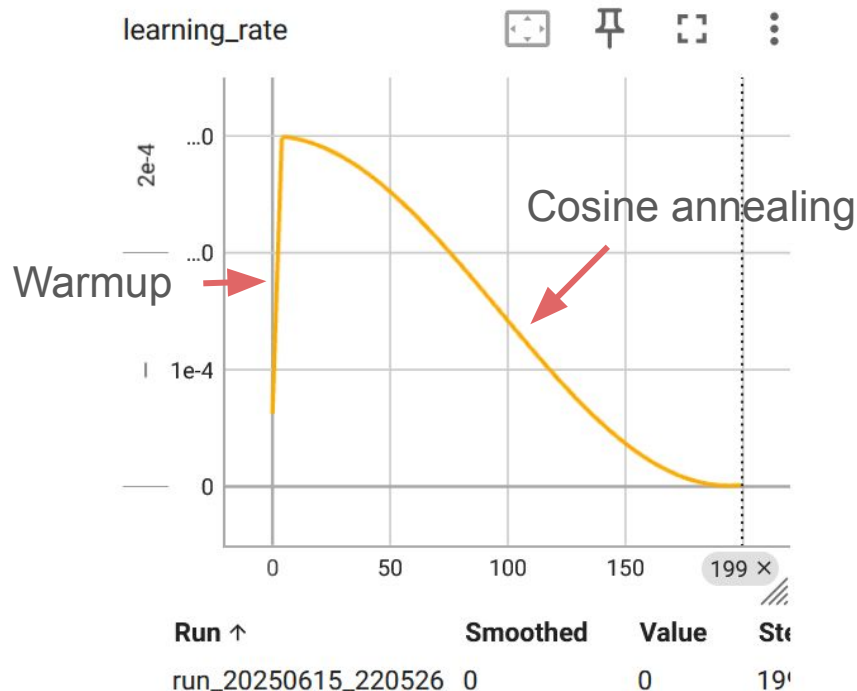
Workflow

1. Read in data
2. Convert label format “0 1 26” -> [1, 1, 0, ..., 0, 1, 0] (Multi-Hot-Labels)
3. Train-Test-Split
4. Create Datasets / DataLoader
5. Modify / Initialize model (Swap input layer of pretrained model)
6. Freeze parts of the model
7. Initialize optimizer / learning-rate-scheduler / loss-function
8. Training loop
9. Test loop

Techniques

1. Data augmentation
2. AdamW optimizer
3. Input layer modification
4. Model freezing
5. Learning rate scheduler: Cosine annealing with warmup
6. (Threshold optimization)

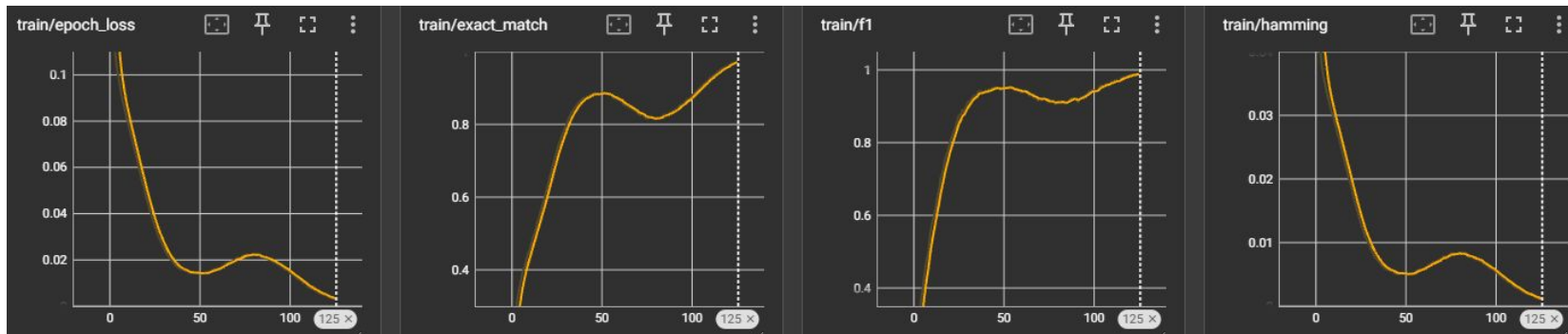
Learning rate scheduler



Cosine annealing with warmup

- Stabilizes training in early stages
- Prevents “damage” to the model

Results



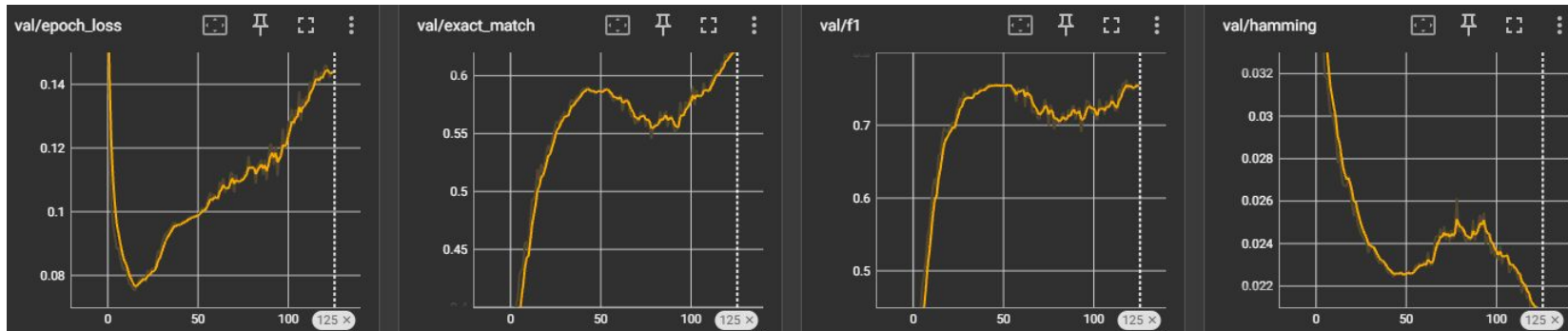
Train
Test

Loss

Accuracy 0.9740
0.6208

Macro F1 0.9901
0.7532

Hamming distance



F1 - Train set



F1 - Validation set



Why is our model overfitting?

1. **Complexity:** Millions of parameters, are a key factor.
2. **Data:** Insufficient training data, contributes to overfitting.
3. **Imbalance:** Our classes were very unevenly distributed, exacerbating overfitting

How to prevent overfitting

1. **Increase Training Data:** Provide more diverse examples to the model.
2. **Simplify Model:** Reduce the number of parameters or layers.
3. **Early Stopping:** Stop training when performance on a validation set degrades.
4. **Regularization:** Add penalties for large weights (L1/L2 regularization, dropout).
5. **Data Augmentation:** Create new training samples by transforming existing ones.
6. **Cross-Validation:** Evaluate model performance on multiple data splits.

Ideas for the future

1. Threshold optimization
2. Use cross validation
3. Tune hyperparameters
4. Train on full sized images 2048x2048 (current 512x512)
5. Train on entire dataset
6. Train model on smaller image patches
7. Try CosineAnnealing with warm restarts
8. Prune model parameters (Prune-Aware-Training)
9. Try different loss functions

Stats

Training time: 140 Hours

Total epochs: 2000

Models parameter count: 21 - 86 Million

Dataset size: 16 GB

Competition dataset images: 124.288

Total “human protein atlas” dataset images: ca. 13 Million

Feedback please!

We would be happy to receive feedback from you

Was there something you liked?

Did you learn something new?

What could be done better?

Do you a tip you share?